Lattice QFT on the Interaction Complex

Abstract

This document constructs a finite and explicit $SU(3)$ lattice gauge structure from the interaction graph generated by the viscous force channel of the interacticle run. No axiom verification is performed. Every definition needed for a later verification paper is introduced and proved here.

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1 Imported facts and notation

Definition 1.1 (External assumptions from `01_convergence.tex`)**.** *Let $d = 3$. Fix kernel parameters*

$$\ell_{\text{visc}} > 0, \qquad \kappa_0 > 0, \qquad \nu_{\text{visc}} \geq 0,$$

and

$$K_\rho(x, y) := \kappa_0 + \exp\left(-\frac{\|x - y\|^2}{2\ell_{\text{visc}}^2}\right), \qquad \rho := \ell_{\text{visc}}.$$

From the kernel assumption in `01_convergence.tex`, the kernel satisfies

$$\kappa_* := \inf_{x,y} K_\rho(x, y) = \kappa_0, \qquad \kappa^* := \sup_{x,y} K_\rho(x, y) = 1 + \kappa_0 < \infty.$$

A recorded trajectory is available at discrete times

$$\mathbb{T} = \{0, 1, \ldots, T\}, \quad T \in \mathbb{N},$$

with finite alive sets $\mathcal{A}(t) \subseteq \{1, \ldots, N\}$ and finite node data

$$x_i(t),\, v_i(t) \in \mathbb{R}^3, \qquad \Phi_i(t) \in \mathbb{R}, \qquad i \in \mathcal{A}(t),\ t \in \mathbb{T}.$$

At each timestep t, $\mathcal{P}_t \subseteq \mathcal{A}(t) \times \mathcal{A}(t) \setminus \{(i, i) : i \in \mathcal{A}(t)\}$ is finite and records the ordered interaction companions used by the algorithm at t.

2 Recorded run and interaction complex

Definition 2.1 (Recorded interacticle run)**.** *The recorded run is the tuple*

$$\mathscr{R} = \Big(T, \{\mathcal{A}(t)\}_{t\in\mathbb{T}}, \{\mathcal{P}_t\}_{t\in\mathbb{T}}, \{x_i(t), v_i(t), \Phi_i(t)\}, \{K_\rho\}\Big).$$

Definition 2.2 (Node set and projections)**.** *For each alive pair (i,t) define a spacetime node $n_{i,t}$ and set*

$$\mathcal{N} := \{n_{i,t} \mid i \in \mathcal{A}(t),\ t \in \mathbb{T}\}.$$

Define spatial and velocity projections

$$\mathbf{x}(n_{i,t}) = x_i(t), \qquad \mathbf{v}(n_{i,t}) = v_i(t).$$

Both maps are well-defined by definition of $\mathcal{N}$.

Definition 2.3 (Oriented edge sets)**.** *Define*

$$E_{\mathrm{CST}} := \{(n_{i,t} \to n_{i,t+1}) : i \in \mathcal{A}(t),\ t = 0, \dots, T-1\},$$
$$E_{\mathrm{IG}} := \{(n_{i,t} \to n_{j,t}) : (i,j) \in \mathcal{P}_t,\ t \in \mathbb{T}\},$$
$$E_{\mathrm{IA}} := \{(n_{i,t+1} \to n_{j,t}) : (i,j) \in \mathcal{P}_t,\ t = 0, \dots, T-1\}.$$

Set

$$\mathcal{E} := E_{\mathrm{CST}} \cup E_{\mathrm{IG}} \cup E_{\mathrm{IA}}.$$

For $e = (u \to v) \in \mathcal{E}$, write $s(e) := u$ and $t(e) := v$, and define $e^{-1} := (t(e) \to s(e))$.

Definition 2.4 (Interaction triangles and plaquettes)**.** *For each $(i,j) \in \mathcal{P}_t$ with $t < T$, define the oriented interaction triangle*

$$\triangle_{ij,t} := (n_{i,t}, n_{i,t+1}, n_{j,t}).$$

Its oriented boundary is

$$\partial\triangle_{ij,t} = e_{\mathrm{CST}}(n_{i,t} \to n_{i,t+1}) + e_{\mathrm{IA}}(n_{i,t+1} \to n_{j,t}) - e_{\mathrm{IG}}(n_{i,t} \to n_{j,t}).$$

If both ordered pairs $(i,j), (j,i) \in \mathcal{P}_t$ with $t < T$, define

$$P_{ij,t} := \triangle_{ij,t} \cup \triangle_{ji,t}.$$

Its boundary is the 4-cycle

$$\partial P_{ij,t} = e_{\mathrm{CST}}(n_{i,t} \to n_{i,t+1}) + e_{\mathrm{IA}}(n_{i,t+1} \to n_{j,t}) + e_{\mathrm{CST}}(n_{j,t} \to n_{j,t+1}) + e_{\mathrm{IA}}(n_{j,t+1} \to n_{i,t}).$$

Set

$$\mathcal{T} := \{\triangle_{ij,t} \mid (i,j) \in \mathcal{P}_t,\ t = 0, \dots, T-1\}, \qquad \mathcal{F} := (\mathcal{N}, \mathcal{E}, \mathcal{T}).$$

Proposition 2.5 (Finiteness and cardinality)**.** *All sets in Definitions 2.1–2.4 are finite. For $m_t := |\mathcal{P}_t|$,*

$$|E_{\mathrm{CST}}| = \sum_{t=0}^{T-1} |\mathcal{A}(t)|, \qquad |E_{\mathrm{IG}}| = \sum_{t=0}^{T} m_t, \qquad |E_{\mathrm{IA}}| = |\mathcal{T}| = \sum_{t=0}^{T-1} m_t.$$

For each $t < T$ with $(i,j) \in \mathcal{P}_t$,

$$\partial P_{ij,t} = \partial\triangle_{ij,t} + \partial\triangle_{ji,t},$$

with the shared edge contribution canceling because

$$e_{\mathrm{IG}}(n_{i,t} \to n_{j,t}) = -\, e_{\mathrm{IG}}(n_{j,t} \to n_{i,t})$$

in the oriented sum.

Proof. Finiteness follows from finiteness of $\mathbb{T}$, all $\mathcal{A}(t)$ and all $\mathcal{P}_t$, together with the set definitions. Cardinality identities are by disjoint union over timesteps and one-to-one correspondence between $\mathcal{P}_t$ and contributions in E_{IG}, E_{IA}, and $\mathcal{T}$. The boundary identity is direct from the explicit formulas in Definition 2.4. □

3 Viscous force from interaction edges

Definition 3.1 (IG degree and pairwise force coefficients)**.** *For each (i,t) define the raw degree and normalized IG coefficients*

$$\deg_{\mathrm{IG}}(i,t) := \sum_{(i,j)\in\mathcal{P}_t} K_\rho(x_i(t), x_j(t)),$$

$$\omega_{ij}^{\mathrm{IG}}(t) := \begin{cases} \dfrac{K_\rho(x_i(t), x_j(t))}{\deg_{\mathrm{IG}}(i,t)}, & (i,j)\in\mathcal{P}_t,\ \deg_{\mathrm{IG}}(i,t)>0, \\ 0, & \textit{otherwise.} \end{cases}$$

Definition 3.2 (Viscous pair force and node force)**.** *For each edge $e = (n_{i,t} \to n_{j,t}) \in E_{\mathrm{IG}}$ define*

$$\mathbf{F}^{\mathrm{visc}}(e) := \nu_{\mathrm{visc}}\omega_{ij}^{\mathrm{IG}}(t)\,(v_j(t) - v_i(t)) \in \mathbb{R}^3.$$

For $x = n_{i,t} \in \mathcal{N}$ define

$$\mathbf{F}^{\mathrm{visc}}(x) := \sum_{\substack{e\in E_{\mathrm{IG}} \\ s(e)=x}} \mathbf{F}^{\mathrm{visc}}(e).$$

If the sum is over an empty set, it is 0.

Lemma 3.3 (Weight properties)**.** *For every $i \in \mathcal{A}(t)$ and $t \in \mathbb{T}$,*

$$\omega_{ij}^{\mathrm{IG}}(t) \ge 0, \qquad \sum_{j:(i,j)\in\mathcal{P}_t} \omega_{ij}^{\mathrm{IG}}(t) = \mathbf{1}_{\{\deg_{\mathrm{IG}}(i,t)>0\}}.$$

Proof. Positivity is immediate from $K_\rho \ge \kappa_* > 0$. If $\deg_{\mathrm{IG}}(i,t) > 0$, summing the definitions over j gives

$$\sum_{j:(i,j)\in\mathcal{P}_t} \omega_{ij}^{\mathrm{IG}}(t) = \frac{1}{\deg_{\mathrm{IG}}(i,t)} \sum_{j:(i,j)\in\mathcal{P}_t} K_\rho(x_i(t), x_j(t)) = 1.$$

If $\deg_{\mathrm{IG}}(i,t) = 0$, every summand in Definition 3.1 is 0 by convention and the sum is 0. □

Lemma 3.4 (Node reconstruction of viscous force)**.** *For every $x = n_{i,t} \in \mathcal{N}$,*

$$\mathbf{F}^{\mathrm{visc}}(x) = \sum_{\substack{e\in\mathcal{E} \\ s(e)=x,\ e\in E_{\mathrm{IG}}}} \mathbf{F}^{\mathrm{visc}}(e).$$

Proof. By Definition of E_{IG}, the summands are exactly the ordered companions $(i,j) \in \mathcal{P}_t$ with source $n_{i,t} = x$. This is the same index set used in Definition 3.2. □

Lemma 3.5 (Orthogonal covariance)**.** *Let $O \in O(3)$ and define transformed data $x_i'(t) = Ox_i(t)$, $v_i'(t) = Ov_i(t)$. Then for each node $x = n_{i,t}$,*

$$\mathbf{F}^{\mathrm{visc}}(x; \textit{transformed data}) = O\,\mathbf{F}^{\mathrm{visc}}(x; \textit{original data}).$$

Proof. Kernel values depend only on $\|x_i - x_j\|$, so

$$K_\rho(Ox_i, Ox_j) = K_\rho(x_i, x_j).$$

Therefore $\omega_{ij}^{\mathrm{IG}}(t)$ is unchanged by orthogonal rotation. For each $e = (n_{i,t} \to n_{j,t})$,

$$\mathbf{F}^{\mathrm{visc}\prime}(e) = \nu_{\mathrm{visc}}\omega_{ij}^{\mathrm{IG}}(t)(Ov_j - Ov_i) = O\,\mathbf{F}^{\mathrm{visc}}(e).$$

Summing over outgoing IG edges at x preserves left multiplication by O. □

4 Color triplets and $SU(3)$ edge transport

Definition 4.1 (Normalized color vectors)**.** *Fix constants $g_3 > 0,\ \ell_0 > 0,\ \hbar_{\mathrm{eff}} > 0,\ m > 0,\ \epsilon_c > 0$. For $x = n_{i,t} \in \mathcal{N}$, define*

$$\tilde{c}_x^{(\alpha)} := F_{\mathrm{visc}}^{(\alpha)}(x)\exp\left(i\frac{m\,v_i^{(\alpha)}(t)\ell_0}{\hbar_{\mathrm{eff}}}\right), \quad \alpha = 1, 2, 3,$$

$$\mathbf{c}_x := \frac{\tilde{\mathbf{c}}_x}{\sqrt{\tilde{\mathbf{c}}_x^\dagger\tilde{\mathbf{c}}_x + \epsilon_c^2}} \in \mathbb{C}^3.$$

Then $\mathbf{c}_x^\dagger\mathbf{c}_x \in [0, 1)$ and $\mathbf{c}_x$ is defined for all $x \in \mathcal{N}$.

Proof. By construction $\epsilon_c > 0$, so $\tilde{\mathbf{c}}_x^\dagger\tilde{\mathbf{c}}_x + \epsilon_c^2 > 0$. Hence the denominator is positive, and

$$\mathbf{c}_x^\dagger\mathbf{c}_x = \frac{\tilde{\mathbf{c}}_x^\dagger\tilde{\mathbf{c}}_x}{\tilde{\mathbf{c}}_x^\dagger\tilde{\mathbf{c}}_x + \epsilon_c^2} \in [0, 1).$$

□

Definition 4.2 (Gell–Mann generators)**.** *Let $\lambda_a \in \mathbb{C}^{3\times 3}$, $a = 1, \ldots, 8$, be*

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},\ \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix},\ \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix},\ \lambda_8 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Each λ_a is Hermitian and $\operatorname{tr}\lambda_a = 0$.

Lemma 4.3 (Gell–Mann identities)**.** *These matrices satisfy*

$$\operatorname{tr}(\lambda_a\lambda_b) = 2\delta_{ab}, \qquad a, b \in \{1, \ldots, 8\}.$$

Proof. Direct matrix multiplication gives diagonal quadratic terms 2 for $a = b$ and zero trace for $a \neq b$. The full check is finite and componentwise. □

Lemma 4.4 (Hermitian traceless span)**.** *If $\alpha_1, \ldots, \alpha_8 \in \mathbb{R}$, then*

$$H := \sum_{a=1}^{8} \alpha_a \lambda_a$$

is Hermitian and trace free.

Proof. Each λ_a is Hermitian and trace free by Definition 4.2. Real linear combinations preserve Hermiticity and zero trace. □

Definition 4.5 (Generator and transport on edges)**.** *For each oriented edge $e = (x \to y) \in \mathcal{E}$ define*

$$\Theta_e := \frac{g_3}{2} \sum_{a=1}^{8} \Re\big(\mathbf{c}_x^\dagger \lambda_a \mathbf{c}_y\big) \lambda_a \in \mathfrak{su}(3),$$

$$U_3(e) := \exp(i\Theta_e) \in SU(3),$$

and $U_3(e^{-1}) := U_3(e)^{-1}$.

Proposition 4.6 (Edge transport is $SU(3)$)**.** *For every $e \in \mathcal{E}$, $U_3(e) \in SU(3)$ and $U_3(e^{-1}) = U_3(e)^{-1}$.*

Proof. Each coefficient $\Re(\mathbf{c}_x^\dagger \lambda_a \mathbf{c}_y)$ is real. By Lemma 4.4, Θ_e is Hermitian and trace free, so $\Theta_e \in \mathfrak{su}(3)$ by definition. Hence $i\Theta_e$ is skew-Hermitian and $\exp(i\Theta_e)$ is unitary. Because $\operatorname{tr}\Theta_e = 0$, $\det U_3(e) = \exp(i \operatorname{tr}\Theta_e) = 1$. The relation for inverse edges is forced by the convention $U_3(e^{-1}) := U_3(e)^{-1}$. □

Remark 4.7. When $x = y$ and e is a self-oriented edge, Θ_e is still well-defined and $U_3(e)$ may be nontrivial.

5 Holonomy and gauge invariance

Definition 5.1 (Path holonomy)**.** *For an oriented path*

$$\gamma = (v_0 \to v_1, \ldots, v_{m-1} \to v_m), \qquad v_k \in \mathcal{N},$$

define

$$U_3(\gamma) := \prod_{k=1}^{m} U_3(v_{k-1} \to v_k),$$

ordered from $k = 1$ to $k = m$. If $v_m = v_0$, γ is a loop and $W_3(\gamma) := \operatorname{tr} U_3(\gamma)$.

Definition 5.2 (Triangle and plaquette holonomy)**.** *For a triangle or plaquette with oriented boundary edges $\partial\triangle$ or ∂P,*

$$U_3(\triangle) := \prod_{e \in \partial\triangle} U_3(e), \qquad U_3(P) := \prod_{e \in \partial P} U_3(e),$$

with the order following the oriented boundary sequence.

Proposition 5.3 (Plaquette factorization)**.** *For each plaquette $P_{ij,t}$,*

$$U_3(P_{ij,t}) = U_3(\triangle_{ij,t})\, U_3(\triangle_{ji,t}).$$

Proof. By definition $\partial P_{ij,t} = \partial\triangle_{ij,t} + \partial\triangle_{ji,t}$, and the two IG edge factors are with opposite orientation. By $U_3(e^{-1}) = U_3(e)^{-1}$ the shared factor reduces to identity. □

Remark 5.4. No additional assumptions are used: the result is direct from edge orientation conventions in Definitions 2.3 and 2.4.

Definition 5.5 (Gauge map)**.** *A gauge map is* $\Omega : \mathcal{N} \to SU(3)$*. For an edge* $e = (x \to y)$ *define*

$$U_3^{\Omega}(e) := \Omega(y)\, U_3(e)\, \Omega(x)^{-1}.$$

Define transformed color vectors by $\mathbf{c}_x^{\Omega} := \Omega(x)\mathbf{c}_x$*.*

Proposition 5.6 (Gauge invariance of Wilson traces)**.** *For any closed loop* $\gamma = (v_0, \ldots, v_m = v_0)$*,*

$$\operatorname{tr}\left(U_3^{\Omega}(\gamma)\right) = \operatorname{tr}\left(U_3(\gamma)\right).$$

Proof. Write $U_3^{\Omega}(\gamma) = \prod_{k=1}^{m} \Omega(v_k) U_3(v_{k-1} \to v_k) \Omega(v_{k-1})^{-1}$. Consecutive factors telescope to

$$U_3^{\Omega}(\gamma) = \Omega(v_0)\, U_3(\gamma)\, \Omega(v_0)^{-1}.$$

Trace cyclicity removes the conjugation. □

Definition 5.7 (Wilson action)**.** *Let*

$$\mathcal{P}_{\square} := \{P_{ij,t} \mid (i,j), (j,i) \in \mathcal{P}_t,\ t = 0, \ldots, T-1\}, \qquad \beta := \frac{6}{g_3^2}.$$

Define

$$S_{\mathrm{W}} := \beta \sum_{P \in \mathcal{P}_{\square}} \left(1 - \frac{1}{3}\Re \operatorname{tr} U_3(P)\right).$$

Proposition 5.8 (Gauge invariance of Wilson action)**.** *For any gauge map* Ω*,* $S_{\mathrm{W}}^{\Omega} = S_{\mathrm{W}}$*.*

Proof. Each term in the sum is invariant by Proposition 5.6. Therefore S_{W} is invariant. □

6 Matter sector on the same complex

Definition 6.1 (Matter fields and actions)**.** *For each node* $x \in \mathcal{N}$ *choose*

$$\psi_x \in \mathbb{C}^3, \qquad \bar{\psi}_x \in (\mathbb{C}^3)^{\dagger}, \qquad \phi_x \in \mathbb{R}.$$

Fix $m_F > 0$*,* $m_\phi > 0$ *and a local potential* $V_\phi : \mathbb{R} \to \mathbb{R}$*. For* $e = (x \to y) \in \mathcal{E}$ *set* $\ell_{xy} := \|\mathbf{x}(y) - \mathbf{x}(x)\| + \eta$ *with* $\eta > 0$*.*

$$S_{\mathrm{F}} := \sum_{e \in E_{\mathrm{CST}} \cup E_{\mathrm{IA}}} \bar{\psi}_{t(e)} \left(\psi_{t(e)} - U_3(e)\psi_{s(e)}\right) + m_F \sum_{x \in \mathcal{N}} \bar{\psi}_x \psi_x,$$

$$S_\phi := \frac{1}{2} \sum_{e \in E_{\mathrm{CST}} \cup E_{\mathrm{IG}}} \frac{(\phi_{t(e)} - \phi_{s(e)})^2}{\ell_{xy}} + \sum_{x \in \mathcal{N}} \Big(\frac{m_\phi^2}{2}\phi_x^2 + V_\phi(\phi_x)\Big).$$

Gauge action on matter is

$$\psi_x \mapsto \Omega(x)\psi_x, \qquad \bar{\psi}_x \mapsto \bar{\psi}_x \Omega(x)^{-1}, \qquad \phi_x \mapsto \phi_x.$$

Proposition 6.2 (Matter gauge invariance)**.** *Under this gauge action,*

$$S_{\mathrm{F}}^{\Omega} = S_{\mathrm{F}}, \qquad S_{\phi}^{\Omega} = S_{\phi}.$$

Proof. For each edge term,

$$\bar{\psi}_{t(e)} \left(\psi_{t(e)} - U_3(e)\psi_{s(e)}\right) \mapsto \bar{\psi}_{t(e)}\Omega(t(e))^{-1} \left(\Omega(t(e))\psi_{t(e)} - \Omega(t(e))U_3(e)\Omega(s(e))^{-1}\Omega(s(e))\psi_{s(e)}\right),$$

which is equal to the original term. The mass term $\bar{\psi}_x\psi_x$ is invariant because $\Omega^{-1}\Omega = I$ pointwise. S_ϕ is built from scalars and is unchanged. □

Remark 6.3. Only $E_{\mathrm{CST}} \cup E_{\mathrm{IA}}$ and $E_{\mathrm{CST}} \cup E_{\mathrm{IG}}$ appear in the matter actions. This choice keeps transport and scalar stiffness aligned with the interaction-complex geometry.

7 Finite lattice-QFT data package

Theorem 7.1 (Finite $SU(3)$ data object)**.** *Given a run as in Definition 2.1 and the constructions in Sections 2–6, the finite tuple*

$$\mathcal{Q} = \Big(T, \{\mathcal{A}(t)\}, \{\mathcal{P}_t\}, \mathcal{N}, \mathcal{E}, \mathcal{T}, \{x_i(t), v_i(t), \Phi_i(t)\}_{i,t}, K_\rho, \nu_{\mathrm{visc}}, \hbar_{eff}, m, \ell_0, \epsilon_c, g_3, \beta, \eta, \{U_3(e)\}_{e\in\mathcal{E}}, S_{\mathrm{W}}, S_{\mathrm{F}}, S_\phi\Big)$$

is explicit and finite. It carries:

1. *the directed interaction complex* $(\mathcal{N}, \mathcal{E}, \mathcal{T})$*;*
2. $SU(3)$ *transport maps on edges;*
3. *Wilson traces and Wilson action;*
4. *gauge action on fermionic and scalar sectors.*

All objects needed for a subsequent axiom-verification paper are available.

Proof. Finiteness of all sets and sums follows from Proposition 2.5 and the finiteness of the run data. All remaining constructions are finite products and sums in finite dimensions. □